We want to try a different way, though: this is clearly an optimization problem, so why not trying to tackle it with our new shining tool, simulated annealing?

A word of caution is necessary here: we have mentioned Maslow's hammer law a few times across this book, and this configures like a case where we need to think carefully before choosing which tool to employ. The risk, as you should know by now, is being tempted to use a hammer (simulated annealing) when a screwdriver could work better.
So, before deciding if simulated annealing is worth implementing, we would need to ask ourselves a few questions, like: do we have the skills, inside our team/company, to develop and tune this algorithm? Would we have better skills for another solution? What's the difference in effort needed between these solutions? What's the benefit of one versus the other?

Simulated annealing can potentially bring us a better solution than the average provided by the MST heuristic, it can even lead us to global optimum. Moreover, let's assume we don't have expertise on the specific problem inside the company: overall, it might be worth trying simulated annealing, which is high-level and potentially could be reused for other optimization problems in the future.

When it comes to the cost function, we are in luck: it naturally stems from the problem's very definition, it's the